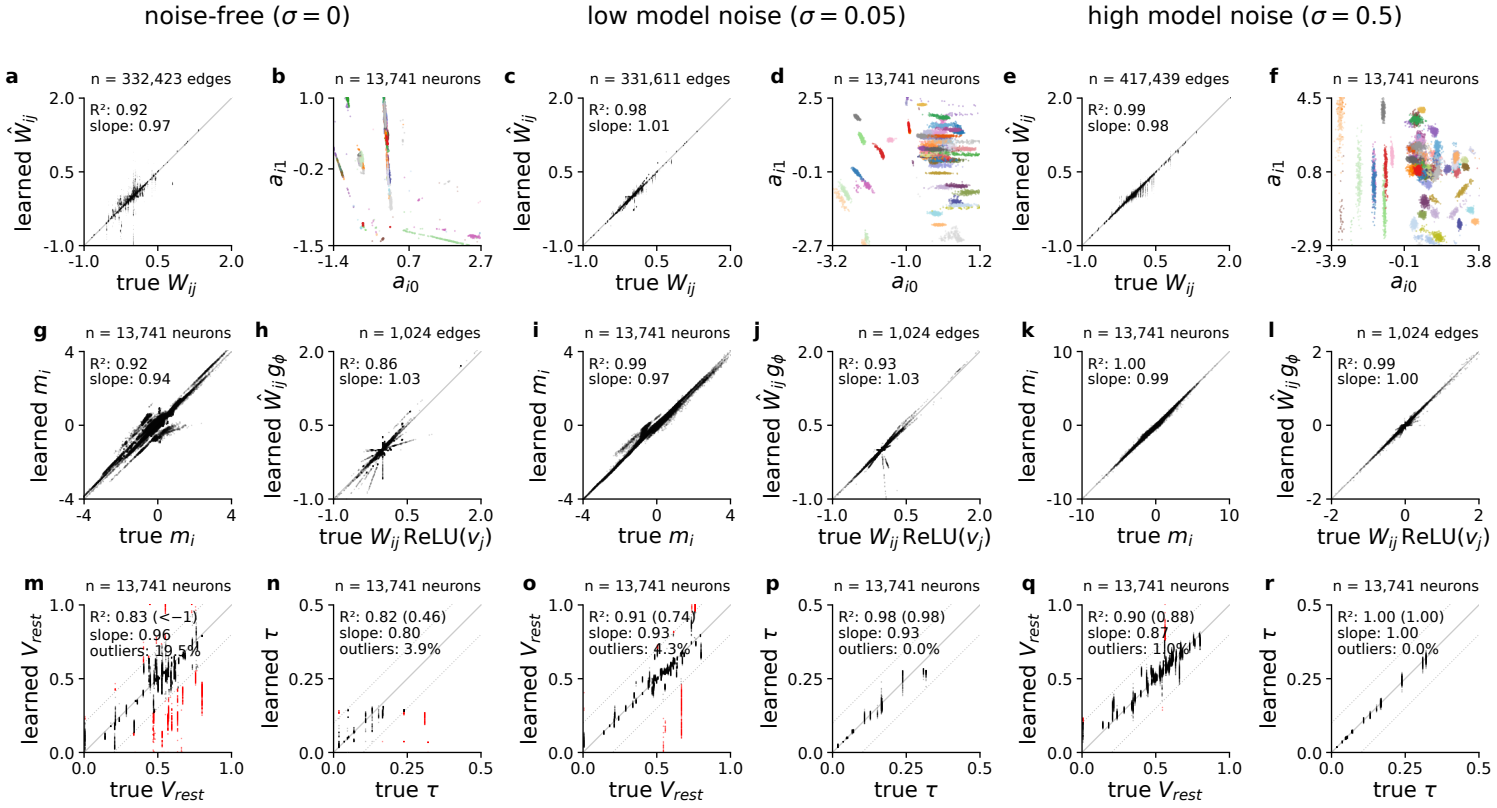

noise-free ($\sigma = 0$)
low model noise ($\sigma = 0.05$)
high model noise ($\sigma = 0.5$)
a
n = 332,423 edges
R²: 0.92
slope: 0.97
learned $\hat{W}_{ij}$
true W_{ij}
b
n = 13,741 neurons
a_{i1}
a_{i0}
c
n = 331,611 edges
R²: 0.98
slope: 1.01
learned $\hat{W}_{ij}$
true W_{ij}
d
n = 13,741 neurons
a_{i1}
a_{i0}
e
n = 417,439 edges
R²: 0.99
slope: 0.98
learned $\hat{W}_{ij}$
true W_{ij}
f
n = 13,741 neurons
a_{i1}
a_{i0}
g
n = 13,741 neurons
R²: 0.92
slope: 0.94
learned m_i
true m_i
h
n = 1,024 edges
R²: 0.86
slope: 1.03
learned $\hat{W}_{ij}\, g_\phi$
true $W_{ij}\,\mathrm{ReLU}(v_j)$
i
n = 13,741 neurons
R²: 0.99
slope: 0.97
learned m_i
true m_i
j
n = 1,024 edges
R²: 0.93
slope: 1.03
learned $\hat{W}_{ij}\, g_\phi$
true $W_{ij}\,\mathrm{ReLU}(v_j)$
k
n = 13,741 neurons
R²: 1.00
slope: 0.99
learned m_i
true m_i
l
n = 1,024 edges
R²: 0.99
slope: 1.00
learned $\hat{W}_{ij}\, g_\phi$
true $W_{ij}\,\mathrm{ReLU}(v_j)$
m
n = 13,741 neurons
R²: 0.83 (<−1)
slope: 0.96
outliers: 19.5%
learned V_{rest}
true V_{rest}
n
n = 13,741 neurons
R²: 0.82 (0.46)
slope: 0.80
outliers: 3.9%
learned τ
true τ
o
n = 13,741 neurons
R²: 0.91 (0.74)
slope: 0.93
outliers: 4.3%
learned V_{rest}
true V_{rest}
p
n = 13,741 neurons
R²: 0.98 (0.98)
slope: 0.93
outliers: 0.0%
learned τ
true τ
q
n = 13,741 neurons
R²: 0.90 (0.88)
slope: 0.87
outliers: 1.0%
learned V_{rest}
true V_{rest}
r
n = 13,741 neurons
R²: 1.00 (1.00)
slope: 1.00
outliers: 0.0%
learned τ
true τ